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> [cs](#) > arXiv:2405.15461

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[Submitted on 24 May 2024 ([v1](#)), last revised 9 Aug 2024 (this version, v5)]

Optimal market-neutral currency trading on the cryptocurrency platform

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This research proposes a novel arbitrage approach in multivariate pair trading, termed the Optimal Trading Technique (OTT). We present a method for selectively forming a "bucket" of fiat currencies anchored to cryptocurrency for monitoring and exploiting trading opportunities simultaneously. To address quantitative conflicts from multiple trading signals, a novel bi-objective convex optimization formulation is designed to balance investor preferences between profitability and risk tolerance. We understand that cryptocurrencies carry significant financial risks. Therefore this process includes tunable parameters such as volatility penalties and action thresholds. In experiments conducted in the cryptocurrency market from 2020 to 2022, which encompassed a vigorous bull run followed by a bear run, the OTT achieved an annualized profit of 15.49%. Additionally, supplementary experiments detailed

in the appendix extend the applicability of OTT to other major cryptocurrencies in the post-COVID period, validating the model's robustness and effectiveness in various market conditions. The arbitrage operation offers a new perspective on trading, without requiring external shorting or holding the intermediate during the arbitrage period. As a note of caution, this study acknowledges the high-risk nature of cryptocurrency investments, which can be subject to significant volatility and potential loss.

Comments: 24 pages, 8 figures, 11 tables

Subjects: **Computational Engineering, Finance, and Science (cs.CE)**; Mathematical Finance (q-fin.MF)

MSC classes: 91B28, 62P05

Cite as: [arXiv:2405.15461](https://arxiv.org/abs/2405.15461) [cs.CE]
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Submission history

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[\[v1\]](#) Fri, 24 May 2024 11:35:49 UTC (934 KB)

[\[v2\]](#) Sat, 1 Jun 2024 04:06:59 UTC (934 KB)

[\[v3\]](#) Tue, 23 Jul 2024 00:19:24 UTC (973 KB)

[\[v4\]](#) Thu, 25 Jul 2024 00:30:00 UTC (973 KB)

[\[v5\]](#) Fri, 9 Aug 2024 11:29:41 UTC (2,049 KB)

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